

Cynthia Cesar

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EDUCATION

Institution: Miami Dade College Kendall Campus

Kendall, FL

Associate of Arts in Electrical Engineering

Completed June 2025

Leadership: Vice President of MDC Kendall Society of Women Engineers (SWE) Chapter, IEEE member, and Introductory Epoxy Metal Polishing Research

TECHNICAL SKILLS

CAD Modeling (Proficient in SolidWorks, Intermediate in Fusion 360); **Programming** (Python and C++); **3D Printing; Lab Equipment** (Multimeter, Oscilloscope, DC Power Supply, Buehler EcoMet 30); Soldering, Circuit Analysis (Direct Current and Digital Logic)

WORK EXPERIENCE

Robotics & Coding Instructor - Bricks 4 Kidz | Tekniklabs

Kendall, FL | *May 2026 - August 2026*

- Designed and delivered hands-on STEM activities introducing elementary students to robotics, programming, electronics, and engineering-based problem solving.
- Guided students through LEGO robotics projects, introductory programming, Minecraft Java/Roblox scripting, and basic circuit-design activities.
- Adapted technical instruction and project difficulty to different age groups and skill levels, translating engineering concepts into practical, interactive challenges.
- Encouraged students to troubleshoot failed designs, test different solutions, and iterate on their projects rather than relying on predetermined solutions.

FedEx Office - Retail & Shipping Associate

Doral, FL | *November 2025 - May 2026*

- Processed domestic and international shipments by weighing and measuring packages, preparing shipping labels, and handling required customer and shipment information.
- Assisted customers with printing, copying, scanning, faxing, and document-preparation services, including helping troubleshoot issues with printers and other equipment.
- Managed multiple customer orders and service requests during high-volume periods while maintaining accuracy and meeting customer deadlines.
- Supported print production by preparing, processing, and completing customer print orders, including document binding, cutting, finishing, and other production services while meeting quality and turnaround expectations.

Branches, Inc. – Summer Shade Guide / Camp Counselor

Florida City, FL | *Summer 2024 - Summer 2025*

- Supervised and supported kindergarten and first-grade students throughout daily summer programming, maintaining a safe, organized, and welcoming environment.
- Planned and led activities for students, including educational lessons, food-related activities, and creative group games.
- Created and taught new activities based on students' interests, including opportunities for learning, creativity, teamwork, and personal development.
- Maintained the program space throughout the day by washing dishes, sweeping, organizing activity areas, putting away supplies, and cleaning up after activities.

CAMPUS & COMMUNITY INVOLVEMENT

Institute of Electrical and Electronics Engineers (IEEE) | Miami Dade College, Kendall Campus

August 2025 - Present

- Support faculty-led research in mechanical prototyping, precision polishing, and materials finishing, enhancing surface quality, dimensional accuracy, and mechanical performance.
- Apply advanced lab techniques in materials characterization and component testing, gaining hands-on experience in mechanical and electrical engineering applications.

Organic Gardening Volunteer | Miami Dade College, Kendall Campus

2026 - Present

- Volunteer with the Miami Dade College Organic Gardening program, helping maintain the campus vegetable garden while learning about sustainable, chemical-free growing practices.
- Assist with hands-on garden maintenance, including planting seasonal crops, weeding, mulching, composting, and watering to support healthy plant growth and maintain the garden space.
- Contribute to keeping the garden clean, organized, and accessible while working alongside other volunteers during recurring garden workdays.
- Gain practical experience in organic gardening and environmental sustainability while contributing to a campus resource that promotes hands-on learning, community involvement, and responsible care for the environment.

PROJECTS

Smart Jack-in-the-Box | CircuitBox Audio Amplification Systems

September 2025 – October 2025

- Engineered an Arduino-driven interactive jack-in-the-box integrating a DS3235 32.768 kHz servo motor, ultrasonic sensor, LED signaling, and custom audio amplification. Designed and soldered the complete wiring schematic, tested multiple speaker configurations to improve audio output, and debugged electrical connections throughout the build. Constructed a magnetically retracting wooden enclosure to coordinate the mechanical movement of the box with the electronic control system, demonstrating hands-on experience in circuit assembly, soldering, embedded control, and mechanical fabrication.

Biomimetic Vision & 3D Tracking System

August 2025 - December 2025

- Engineered a modular biomimetic vision system integrating a Raspberry Pi 4-based 3D environmental tracking platform with a servo-actuated animatronic eye mechanism. Developed real-time computer vision algorithms for object detection and dynamic spatial tracking, enabling the system to identify and follow individuals within an indoor environment. Designed and fabricated a 3D-printed pan-tilt eye assembly powered by MG90S servo motors and controlled through embedded microcontroller logic to replicate smooth, human-like gaze behavior and synchronized eyelid motion. Separated sensing and actuation architectures to improve modularity, allowing the Raspberry Pi to handle high-level visual processing while the microcontroller executes precise PWM-based motor control. The system served as a scalable foundation for research in robotics, expressive animatronics, and human-machine interaction.

BCD 24-Hour Digital Clock

March 2026 - May 2026

- Individually designed and innovated a fully synchronous 24-hour Binary-Coded Decimal (BCD) clock, while independently designing and building my own complete unit. Implemented a precise 00:00:00–23:59:59 timekeeping architecture using a 32.768 kHz crystal oscillator with CD4060 and cascaded CD4518 devices, integrating combinational reset logic to enforce accurate 60-second, 60-minute, and 24-hour rollover with adaptable reset functionality. Incorporated a gated time-set control that pauses counting for manual adjustment of hours, minutes, and seconds. Advanced from breadboard prototyping to a soldered perf-board build, with planned PCB fabrication and mechanical enclosure design, demonstrating full electrical and mechanical system integration.

FPV Drone

2026 - Present

- Learning to build an FPV drone from the ground up using custom carbon fiber, working with components such as brushless motors, ESCs, flight controllers, radio receivers, FPV cameras, and power distribution.
- Developing hands-on exposure with soldering, wiring, electronics troubleshooting, flight-controller configuration, and basic flight tuning through iterative testing.
- Learning how the drone's flight, control, power, and video systems communicate and work together while diagnosing issues that arise during assembly and testing.
- Exploring future applications of the platform through Raspberry Pi integration, computer vision, object tracking, sensor integration, and onboard data processing.